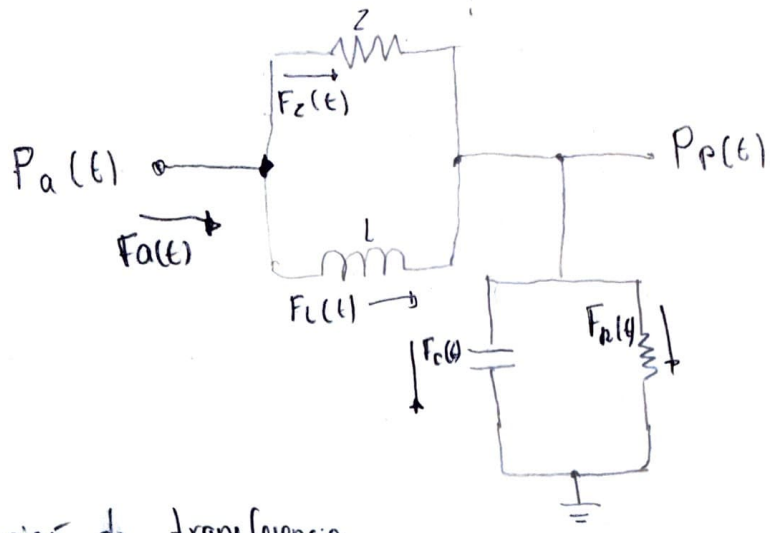


Función de transferencia

Ecuaciones principales

$$Fa(t) = Fz(t) + FL(t) = Fc(t) + Fz(t)$$

$$Fz(t) = \frac{Pa(t) - Pp(t)}{Z}$$

$$FL(t) = \frac{1}{L}\int [Pa(t) - Pp(t)]\,dt$$

$$Fc(t) = C\frac{dPp(t)}{dt}$$

$$Fe(t) = \frac{Pp(t)}{R}$$

Transformada de laplace

$$Fa(s) = Fz(s) + FL(s) = Fc(s) + FR(s)$$

$$Fz(s) = \frac{Pa(s) - Pp(s)}{Z}$$

$$FL(s) = \frac{Pa(s) - Pp(s)}{Ls}$$

$$Fc(s) = Cs\,Pp(s)$$

$$FR(s) = \frac{Pp(s)}{R}$$

$$\frac{Pa(s) - Pp(s)}{2} + \frac{Pa(s) + Pp(s)}{Ls} = CsPp(s) + \frac{Pp(s)}{R}$$

Procedimiento

$$F_2(s) + F_1(s) = F_3(s) + F_4(s)$$

$$\frac{P_a(s) - P_p(s)}{2} + \frac{P_a(s) - P_p(s)}{Ls} = C_s P_p(s) + \frac{P_p(s)}{R} \qquad e(t) = 0V$$

$$\frac{P_a(s)}{2} - \frac{P_p(s)}{2} + \frac{P_a(s)}{Ls} - \frac{P_p(s)}{Ls} = C_s P_p(s) + \frac{P_p(s)}{R}$$

$$P_a(s)\left(\frac{1}{2} + \frac{1}{Ls}\right) = P_p(s)\left(C_s + \frac{1}{R} + \frac{1}{2} + \frac{1}{Ls}\right)$$

$$P_a(s)\,\frac{Ls+2}{Ls2} = P_p(s)\,\frac{CLR2s^2 + L(2+R)s + R2}{RLZs} \qquad \text{Resultado 2}$$

$$\frac{P_p(s)}{P_a(s)} = \frac{\frac{Ls+2}{Ls2}}{\frac{CLR2s^2 + L(2+R)s + R2}{RLZs}} = \frac{P_p(s)}{P_a(s)} = \frac{LRs + R2}{CLR2s^2 + L(2+R)s + R2}$$

Estabilidad en lazo abierto

$$CLR2s^2 + L(2+R)s + R2 \qquad a = CLR2 \quad b = L(2+R) \quad c = R2$$

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-L(2+R) \pm \sqrt{L^2(2+R)^2 - 4(CLR2)(R2)}}{2CLR2}$$

Hipotenso

$Z = 0.010$
$L = 0.001$
$R = 1.00$
$C = 3.50$

$$\frac{P_p(s)}{P_a(s)} = \frac{LRs + R2}{CLR2s^2 + L(2+R)s + R2} = \frac{0.001s + 0.01}{3.5\times10^{-5}s^2 + 0.00101s + 0.01}$$

$\lambda_1 = -14.42 + 8.86i$
$\lambda_2 = -14.419 - 8.86i$

Presenta una respuesta estable y subamortiguada

Normotenso

$Z = 6.010$
$L = 0.001$
$R = 1.00$
$C = 1.50$

$$\frac{P_p(s)}{P_a(s)} = \frac{LR + R2}{CLR2s^2 + [(2+R)s + R2} = \frac{0.001s + 0.01}{1.5\times10^{-5}s^2 + 0.00101s + 0.01}$$

$\lambda_1 = -12.062$

$\lambda_2 = -55.271$

Presenta una respuesta sobreamortiguada y estable

Apertenio

$Z = 0.010$
$L = 0.001$
$R = 1.00$
$C = 1.00$

$$\frac{P_p(s)}{P_a(s)} = \frac{LRCs + R}{CLRZs^2 + L(Z+R)s + RZ}$$

$$= \frac{0.001s + 0.01}{1\times10^{-5}s^2 + 0.00101s + 0.01}$$

$\lambda_1 = -10$
$\lambda_2 = -10$

La respuesta es estable y criticamente amortiguada

Modelo de ecuaciones integro-diferenciales nota, la unica variable desconocida es la salida del sistma

$$F_Z(t) + F_L(t) = F_C(t) + F_R(t)$$

$$\frac{P_a(t)}{Z} + \frac{1}{L}\int |P_a(t) - P_p(t)|\,dt - C\frac{dP_p(t)}{dt} = \frac{P_p(t)}{R} + \frac{P_p(t)}{(Z)}$$

Nota: P_p no se puede despejar de terminos en los que se esta derivando o integrando

$$P_p(t)\left(\frac{1}{R} + \frac{1}{Z}\right) = \frac{P_a(t)}{Z} + \frac{1}{L}\int[P_a(t) - P_p]\,dt - C\frac{dP_p(t)}{dt}$$

$$P_p(t) = \left[\frac{P_a(t)}{Z} + \frac{1}{L}\int |P_a(t) - P_p(t)|\,dt - C\frac{dP_p(t)}{dt}\right]\frac{RZ}{Z-R}$$